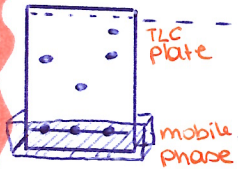


Chromatography Theory

= separation of components based on their different distribution coefficients between mobile & stationary phase!

TLC - Thin layer chromatography

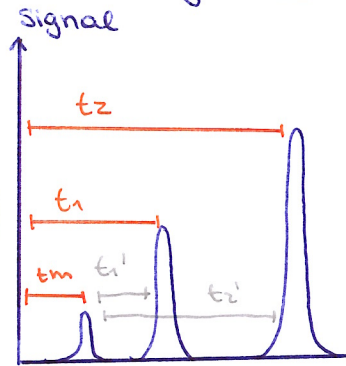


$$R_f = \frac{d(\text{start})}{d(\text{end})} \text{ of mobile phase front}$$

only qualitative !!
helps judge polarity

maybe UV intensity = quantitative

Chromatogramm



t_m = time mobile phase takes to reach end of column (dead time)

t_1, t_2 = retention time

$\alpha = \frac{t_2'}{t_1'}$ = relative retention separation

t_m = time that analyte is in mobile phase

t_1' = time that analyte is on stationary phase

retention volume time, volume or factor can be measured

k = Retention factor = $\frac{C_s}{C_m} = \frac{\text{concentration analyte stat}}{\text{conc. analyte mobile phase}}$

$$k = \frac{t_r - t_m}{t_m} = \frac{t_r'}{t_m}$$

Peak broadening

→ the column is separated into segments (disks / plates)
→ in every segment an equilibrium between mobile & stat. phase is established. Then it flows into the next disk.

→ if we increase the number of disks = equilibrium is more often established = better separation.

→ longer column = more disks (duh.)

→ thickness of disks can be changed (depends on analyte, velocity, etc)

Van Deemter equation:

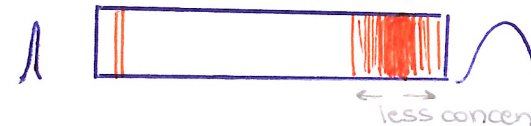
$$\begin{aligned} \text{Height Equivalent Theoretical Plate} &= H_A + H_B + H_C \\ &= A + \frac{B}{v} + C \cdot v \end{aligned}$$

A = Eddy dispersion



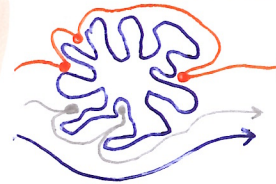
= depends on size of particles
= constant

B = Longitudinal diffusion



$B \cdot v = H_B$ - depends on velocity

C = Mass transfer

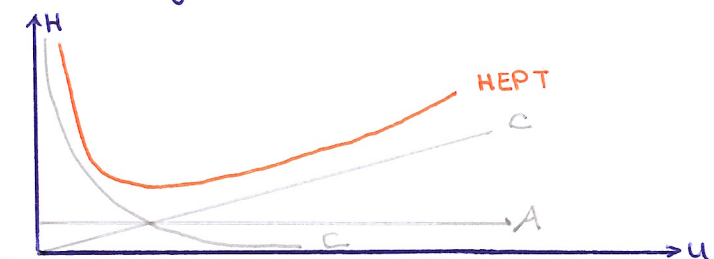


= resistance to mass transfer between mobile and stationary phase

$$H_C = \frac{C}{v}$$

Intotal:

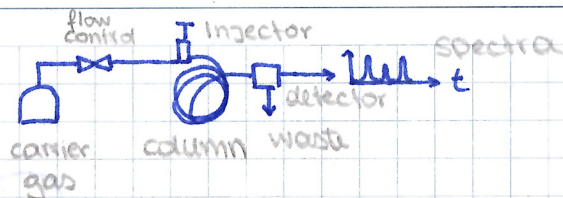
too slow is worse than too fast!



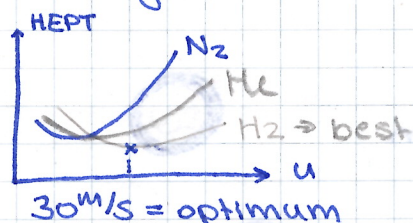
used for small & volatile samples that can be vaporized for pollutants, industrial by products & metabolites of drugs!

1) GC alcohol

GC:



1) carrier gas



purification:

- carbon (for hydrocarbons)
- mol sieves (for H₂O)
- dispersed metallic Cu (for O₂)

- carrier gas MUST NOT react with analyte
- dry & free of oxygen!

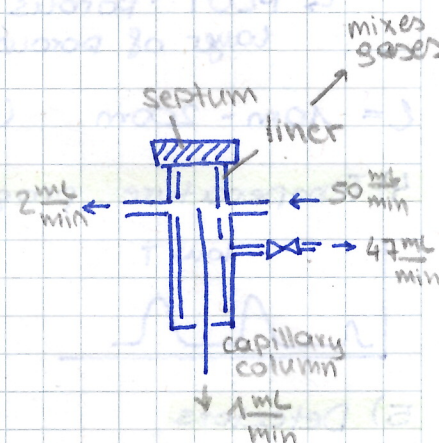
2) Injector a) liquid

→ split / splitless

only portion of the sample (reduces overload)

whole sample for low conc.

depends on sensitivity of the column



- hot injection on hot column ∴ possible fraction
- hot injection on "cold" column (splitless) ☺

→ PTV: Programmable temperature vaporization

→ regulates vaporization

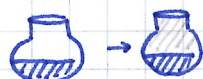
- solvent is evaporated via split valve
- temperature is elevated into column
- * good for traces; no overload; no discrimination

→ cold on column

- directly on column with syringe
- for labile (temperature sensitive) analytes

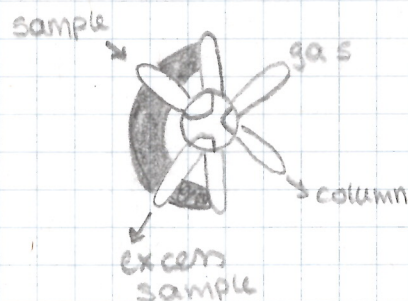


→ Head space



only high vapour pressure

b) gas



to make sure ⇒ no air bubbles!

1) GC

3) column

capillary column:

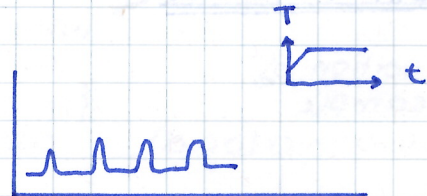
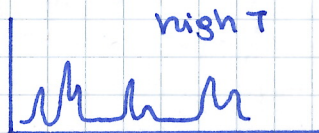
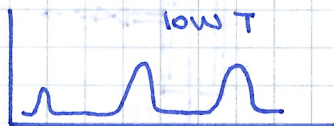
! ↳ WCOT: Wall coated open tubular = most used
thin film of liquid polymer (PDMS, polyethylene)
small sample volume, for traces

↳ SCOT: support coated open tubular
layer of porous particles & on top a film
(alumina, silica, mol sieves)

↳ PLOT: porous layer open tubular
layer of porous particles

$l = 10\text{m} - 200\text{m}$ (usually 30m)

4) Temperature programm



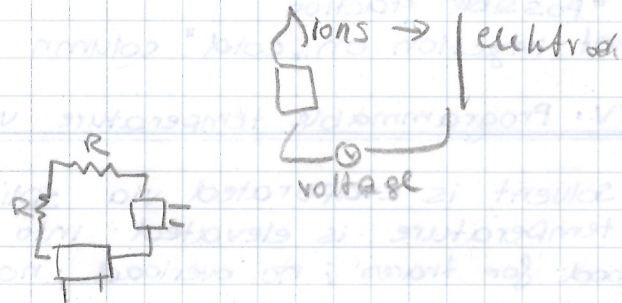
5) Detectors

↳ FID Flame ionization Detector

↳ WLD Wärmeleitfähigkeit

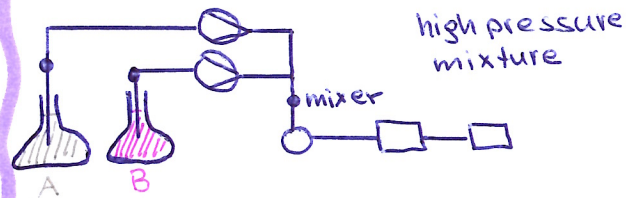
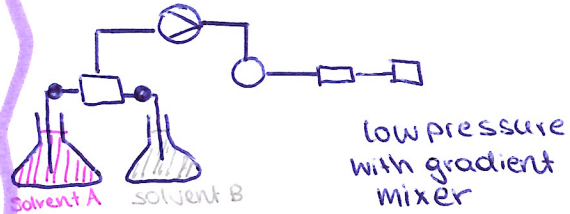
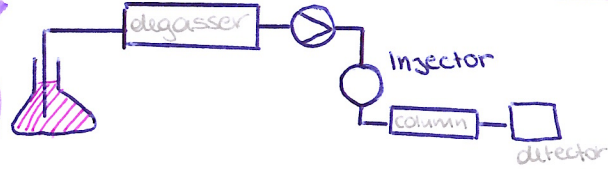
↳ MS

↳ FPD Flame Photometric



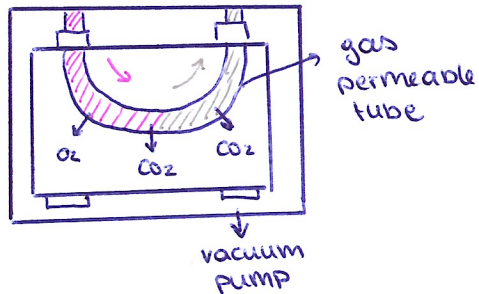
HPLC: High Performance liquid chromatography

Instrument



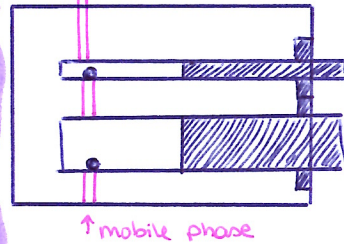
degasser

mobile phase must be degassed to avoid bubbles in the detector



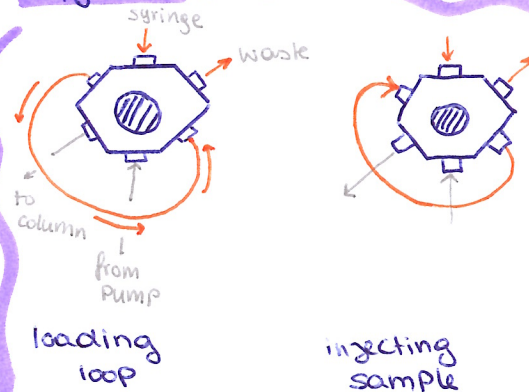
or purge with helium

Pump



- small & large piston are 180° out of phase
- big piston sucks mobile phase in; smaller out

Injector



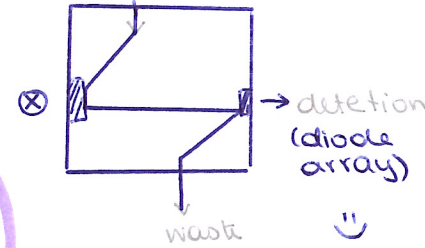
- 1μL - 1mL
- small volume = small column

column:

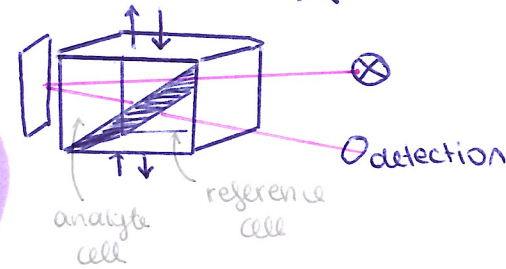
- Packed with particles
- stainless steel
- guard column to protect main column

Detectors

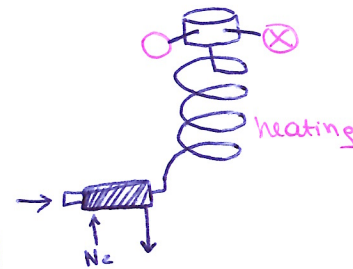
1) UV-vis flow cell



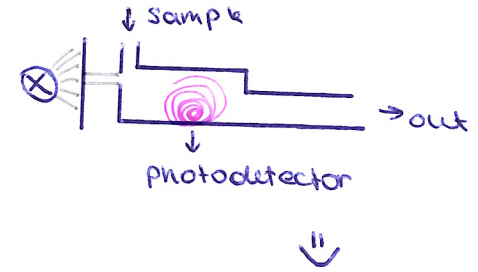
3) Refractive index



4) evaporative light scattering



2) Fluorescence



5) Amperometric

- analyte is ox or red.
- voltage is applied between two electrodes
- current changes when analyte is ox/red.



S/N!
sensitivity
linearity
comparability
conc. dependency

Gradient? UV-Vis, Fluorescence

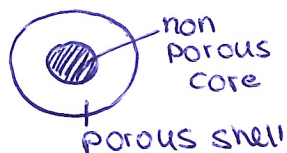
Linear range? UV-Vis - Fluorescence

Selective? Amperometric, Fluorescence

HPLC

Stationary phase

- small particles
= high N
- different porosity



Normal phase NP

stationary = polar
mobile = nonpolar
(Hexane)

- retention increases with increasing polarity of analyte
- decreases with polarity of the mobile phase

eg. silica gel (pH < 8!)
they repel each other at higher pH
alumina: $Al_2O_3 \cdot H_2O$
or polar silanol groups on silica

HILIC: Hydrophilic Interaction Liquid Chromatography
= least polar analyte elutes first

Reverse-phase RP

stationary: nonpolar
mobile: polar

Retention decreases with increasing polarity of analyte
eg. porous graphite
modified silica (M!)
Polystyrene

⇒ endcapping of free silanol groups
TMCS: Trimethylchlorosilane

common mobile phase: water or aq. buffers with org. solvents

Isocratic only one mobile phase

Gradient mobile phase changes during measurement

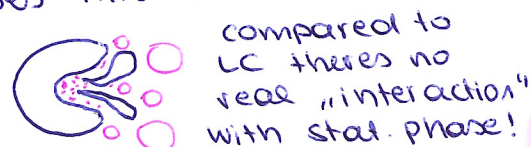
→ mobile phases: NaOH
 $NaHCO_3$

→ After separation a suppressor converts the mobile phase into a low conducting species

Size exclusion chromatography

SEC

= to separate high m.w. molecules
stationary = porous particles (defined pore size)
= Retention difference is controlled by how much the analyte diffuses into the particle



→ stat. phase = silica
→ stat phase = solvent in pores
→ mobile phase = solvent outside

Ion exchange chromatography

= separation of components based on their net charge

$R-N^+(CH_3)_3$ strong anion ex.

$R-NH_3^+$ weak anion ex.

$R-SO_3^-$ strong cat ex.

$R-COO^-$ weak cat ex.

↑ Polymeric resin

Ion exclusion

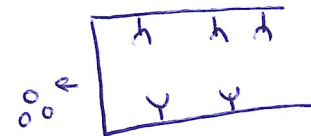
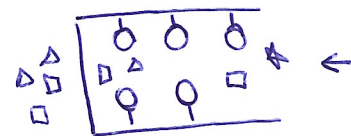
for low molecular weights & some neutral analytes

→ cation exchanger for acids

→ anion exchanger for base

→ mobile phase: H_2O , diluted acid/bases

Affinity

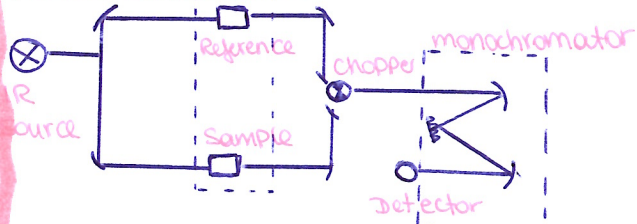


enzyme-substrate
Antigen-Antibody interaction!

IR: Infrared Spectroscopy

Instrument

1) dispersive IR



↳ dual beam to subtract background

2) non-dispersive

Same, but no monochromator

- 1)
 - slow
 - low intensity
 - no constant resolution over λ -range

- 2)
 - + cheaper
 - + fast
 - signals might overlap if close

Radiation source

MIR: ↳ Globar: Silicon carbide



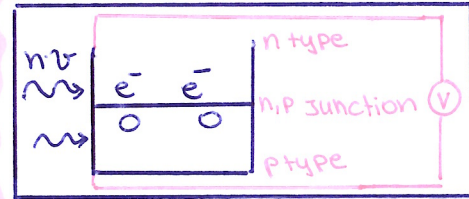
- ↳ low intensity
- ↳ mechanically robust (out door instruments)
- ↳ cooling

↳ Nernst glower:

- ↳ high intensity
- ↳ not robust (only for lab)

NIR: ↳ Quartz-halogen-lamp
↳ Tungsten lamp

2) Photovoltaic detector



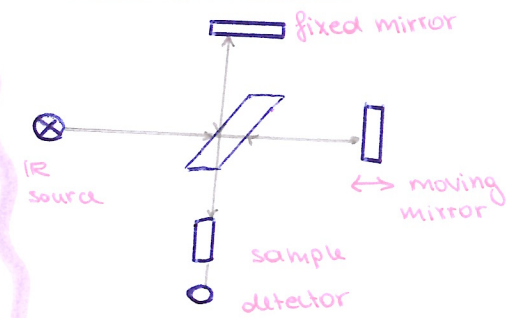
⇒ based on the photoelectric effect
↳ consists of large semiconductors
↳ when exposed to light (IR) mobile charge carriers are created that electronically change the material
(it will become either conductive or cause a photovoltage)

Monochromators

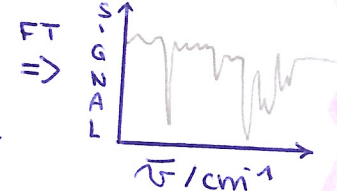
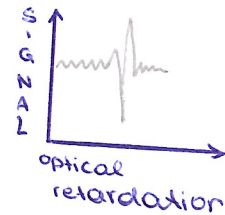
= wave selection

↳ gratings (see left drawing)
often used!

↳ Interferometer: Michelson Interferometer:



• constructive: $m + m = m$
Same phase
• deconstructive $m + m = \text{---}$



- + no sequential scanning for λ
- + constant resolution
- needs FT
- + fast

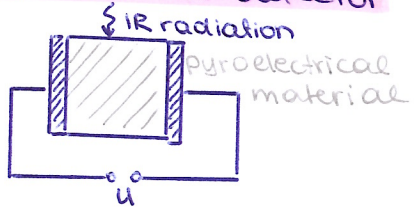
↳ Diode array (for NIR):



$$S/N = \sqrt{n}$$

Detectors

1) Pyroelectrical detector



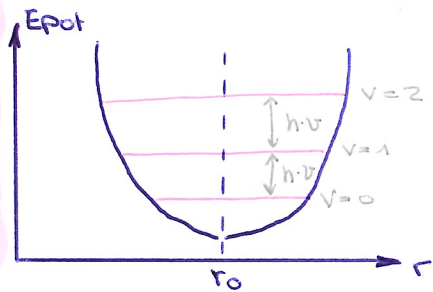
→ ferroelectric crystal or ceramics with an asymmetrical structure ⇒ electric dipole moment & permanent polarization
→ when the material is heated up by IR radiation the distances in lattice change, as well as the polarization!
→ electrical signal
- not sensitive
+ at RT

created that electronically
- has to be cooled with $N_2(l)$
+ high sensitivity

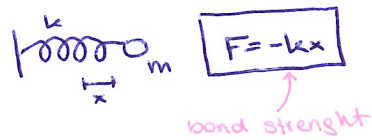
IR: Infrared spectrometer → with Dipole moment

Vibrations

↳ harmonic oscillator

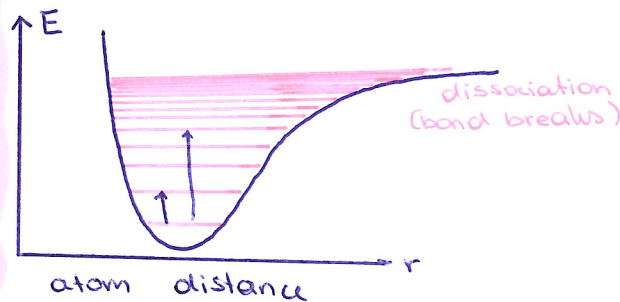


⇒ Hooke's law



$$E = h \cdot \nu_0 \cdot \left(v + \frac{1}{2}\right)$$

↳ non-harmonic oscillator



- bond will break at some point
- distances between levels get smaller
- more transitions possible

Rotations

- In IR we observe a combination of vibrations & rotations
- rotation can only be observed as discrete signals in gas phase. otherwise it shows up as one broad band.

NIR = for structure, qualitative analysis
sometimes used as detector (GC)

specific vibration

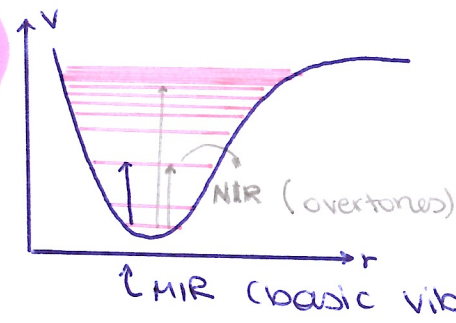
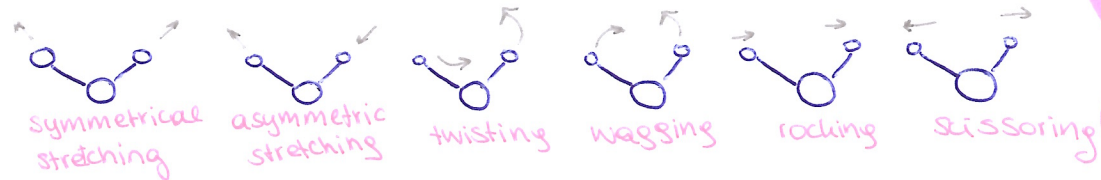
→ selection rule for vibrations:

the dipole moment is important!

It needs to change to be observed!

→ permanent Dipole = FIR

→ temporary Dipole = MIR & NIR



important bands:	
C-C	1000 cm ⁻¹
C=C	1650 cm ⁻¹
C≡C	2200 cm ⁻¹
$\nu = \frac{1}{2\pi} \sqrt{\frac{k}{\mu}}$	

Sampling techniques

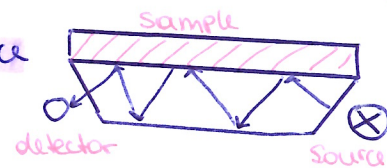
• solids KBr pellets

ATR = Attenuated Total Reflectance

⇒ ZnSe crystal

⇒ radiation intrudes into sample

⇒ works for different shapes



• liquids ATR

in solution with solvent (NOT water! But CHCl₃)

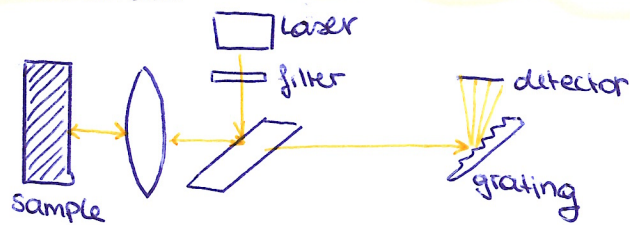
• gases gas cell

NIR = compares with database, quantitative
breath alcohol content

Raman Spectroscopy

= Polarization of a molecule has to change!

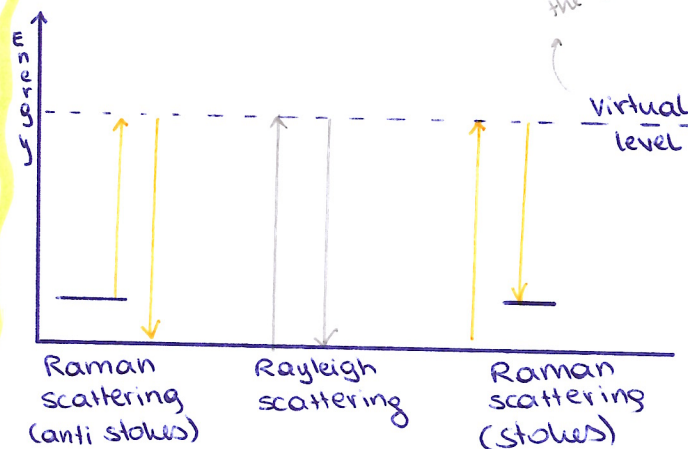
Spectrometer



Source: Hg vapor

Detector: Si Diode

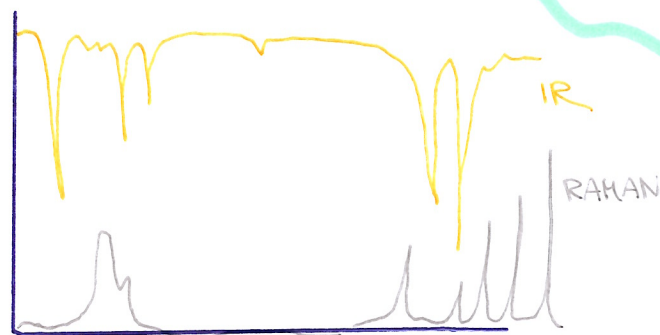
Physical properties:



- Laser light hits the sample
- light mostly passes through
- 10^{-4} elastic Rayleigh scattering
- 10^{-8} Raman scattering
- = these are looked at in the RAMAN spectroscopy! (inelastic scattering)

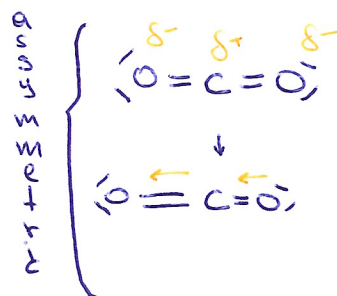
the signal is usually very weak, but can be amplified when using holographic gratings

Spectra



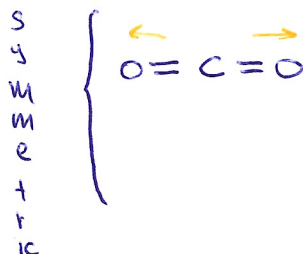
IR & RAMAN complement each other!

another example: CO_2



should be IR inactive, but during vibrations the dipole moment changes

IR active, bc vibration destroys the symmetry
Raman inactive



for a symmetric stretch:

IR inactive

Raman active polarisability changed

bc of symmetry = no dipole

MS - Mass Spectrometry

Ionisation

↳ hard ionization:

lots of energy
lots of fragments

↳ soft ionization:

low energy
nearly no fragments

1) Electron impact ionisation



HARD
VACUUM

EI

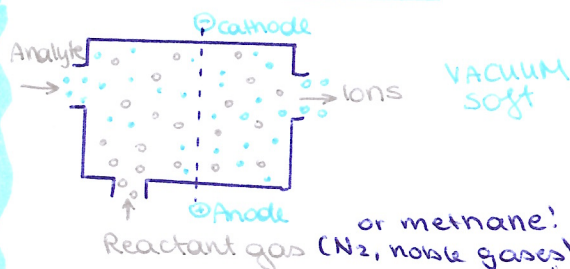
→ e^- are accelerated by an electrical field
~70V but up to 200V, ↑T, ↓p

→ e^- collide with the analyte & kick out
an e^- forming a radical

→ this is unstable & decays into smaller,
charged fragments

+ lots of fragments = info
- only for small molecules

2) Chemical ionisation



VACUUM
soft

CI

→ reactant gas is ionized via the
voltage.

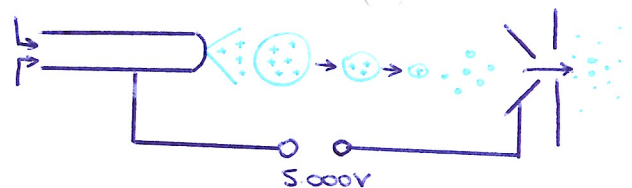
→ The gas ions then react with the
analyte & ionize it.

→ less fragments



3) Electron spray ionization

SOFT 1ATM



for big biomolecules
very sensitive!

• Analyte is in a solvent & a nebulizer gas

• nanodroplets are sprayed & ionized

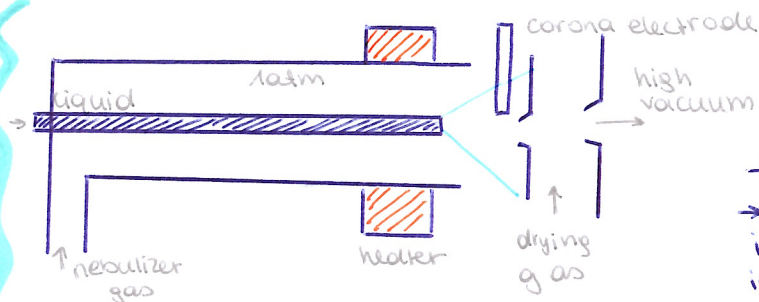
• solvent evaporates = small droplets

+ no fragments

+ no thermal impact

ESI

4) Atmospheric pressure chemical ionization



not for proteins
high heats
(denaturiert)

→ similar to ESI

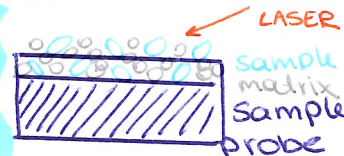
→ but liquid with analyte
is vaporized before
ionization

→ the solvent gets ionized at the electrode

→ for not very polar analytes

APCI

5) Matrix Assisted LASER Desorption Ionization



→ LASER flash produces Matrix neutrals, $\oplus$ & $\ominus$

→ Analyte is ionized by Matrix

MALDI soft (no fragments)

Summary:

want fragments → EI (only small)

want Proteins, DNA? → ESI, MALDI

small & non volatile? → ESI

small & volatile? → EI, APCI, CI

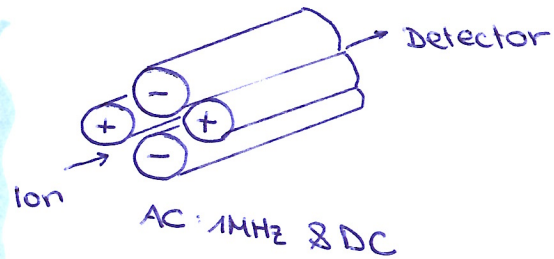
big? → MALDI

MS - Mass Spectrometry

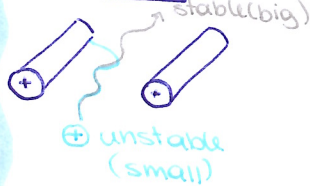
Mass Analyzers

Ions are sorted according to their m/z

1) Quadrupole Mass Analyzer

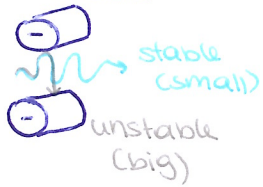


x-direction:



light particles follow the AC field & are lost

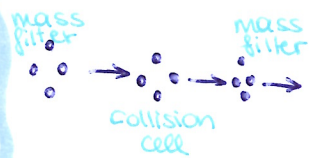
y-direction:



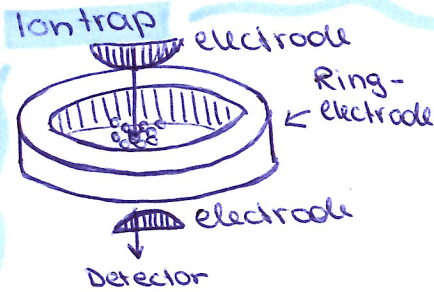
big ones are heavy and stay at the electrode

- + only selected m/z pass
- low accuracy & resolution
- restricted mass range

→ Triple Quadrupole

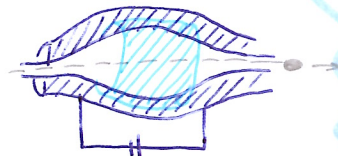


+ very sensitive



- high frequency AC traps the analyte ions in the ring electrode.
- Then ions are freed (lightest first) in resonance with the frequency

Orbitrap



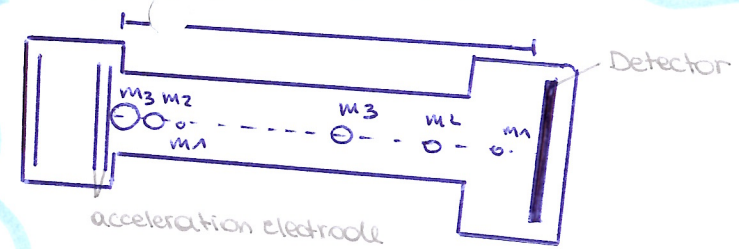
Ions go into the trap with a high velocity through a electrical field. Into a circular motion forced by the $\ominus$ charged inner electrode = axial oscillation

Summary

Quadrupole $\Rightarrow$ triple: for specific fragments very sensitive

Ion trap $\Rightarrow$ more sensitive than Quadrupole
space-charge-effect: unsuitable for quantitative measurement

Time of flight (TOF)

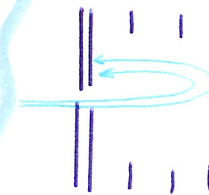


Ions are accelerated with a voltage until all have the same kinetic energy:

$$E_{kin} = \frac{m \cdot v^2}{2} \rightarrow v = \frac{e}{m} \rightarrow t = e \sqrt{\frac{m}{2} \cdot \frac{1}{2V}}$$

$\Rightarrow$ different masses have different velocity (light = faster)

for better resolution: reflection:



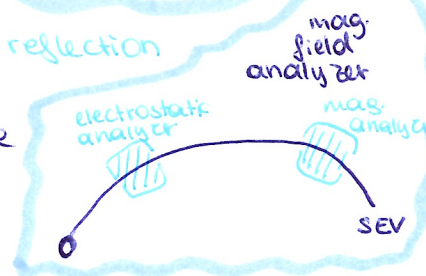
Slower ion = shorter path
Faster ion = longer path

This compensates differences in E_{kin} of ions with same m/z

multiple reflections for even better resolution

TOF $\Rightarrow$ only with reflection

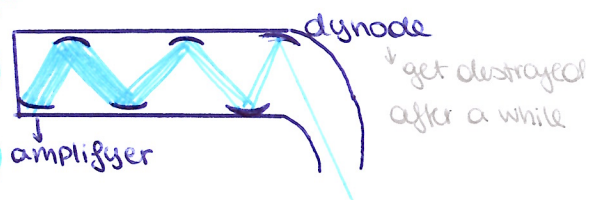
methods can be combined!!



Mass spectrometry

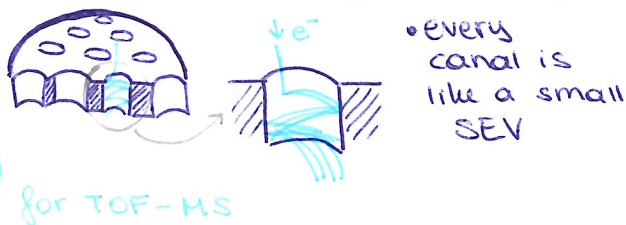
Detector

1) secondary electron multiplier SEV

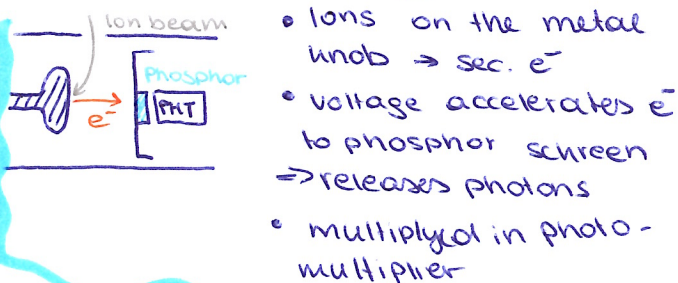


- conversion dynode produces e^- from ions
 - these are accelerated to the dynodes & kick out more e^-
 - produces an electrical signal
- for: Quadrupol, Ion Traps, mag. field

2) Micro channel plate

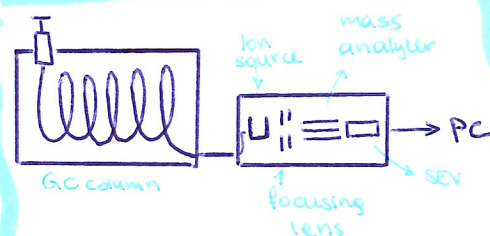


3) Daly Detector



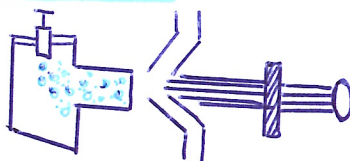
Hyphenation (coupling)

1) GC-MS



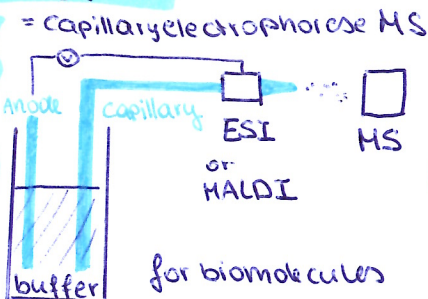
- often with EI/CI (easy vacuum by pumping gaseous mobile phase)
- Quadrupol or TOF for drug detection, food analysis, explosives

2) HPLC-MS



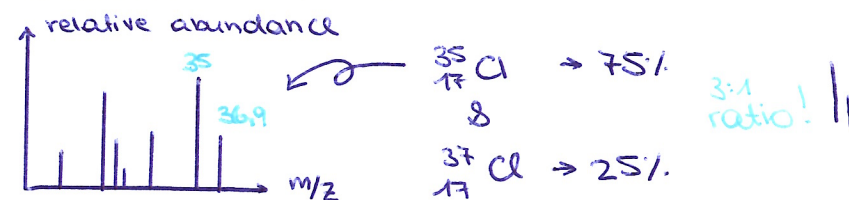
- only with APCI bc no vacuum with liquid
- volatile mobile phase
- any mass analyzer! fast $\rightarrow$ TOF!

3) CE-MS



Spectral data

$\rightarrow$ MS shows isotopic patterns



^{79}Br & ^{81}Br 1:1 ratio !!

$\rightarrow$ you can tell the element by how the peaks look & the ratio!

HRMS=High Resolution MS

$\rightarrow$ up to 0,000u!!

$\Rightarrow$ TOF & Orbitrap

types of masses

1) Nominal mass: $\text{CH}_4 = 12\text{u} + 4 \cdot 1\text{u} = 16\text{u}$
sum of the elements

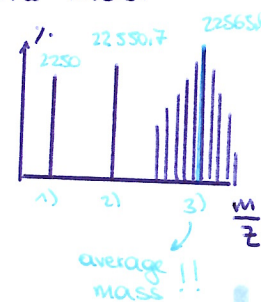
2) mono isotopic mass: exact masses of the most abundant isotope summed up:

$$\text{CH}_4 = 12,000\text{u} + 4 \cdot 1,0078\text{u} = 16,031\text{u}$$

3) molar mass: exact atomic masses

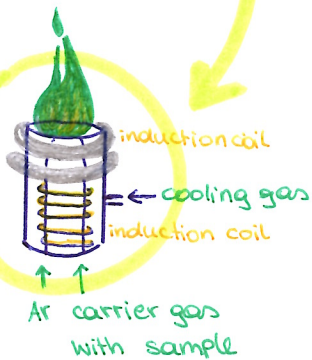
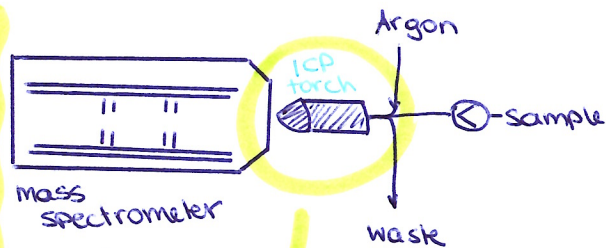
$$\text{CH}_4 = 12,011\text{u} + 4 \cdot 1,0079\text{u} = 16,049\text{u}$$

$$\text{Error} = \frac{m/z_{\text{exp}} - m/z_{\text{theo}}}{m/z_{\text{theo}}} \cdot 10^6 = [\text{ppm}]$$



ICP Inductively coupled plasma

Instrument



- the coil causes a strong magnetic field inside of the torch!!
- torch is quartz and ionizes the sample for the MS!

- free e^- in the RF field (radiofrequency) are accelerated & ionize argon gas
- This then ionizes sample

Detection

1) MS $\rightarrow$ ICP-MS

- very good sensitivity
- few interferences
- multielement
- expensive

2) OES $\rightarrow$ ICPOES

= optical emission

- cheaper
- multielement
- high volume required



Interferences of ICP-MS

$\rightarrow$ isobaric interference:

low resolution instrument can't distinguish between isotopes

$\rightarrow$ molecular interference:

O, H, N, C, Cl NOT Li, Be, B

recombine with Argon =

Don't use HCl bc ArCl

interference $\Rightarrow$ HNO₃ instead.

slightly acidic = better

(to keep analyte in solution before being nebulized!)

The acid conc. has to be the same in the sample and in the STD to stabilize the plasma.

$\rightarrow$ use HR-ICP-MS

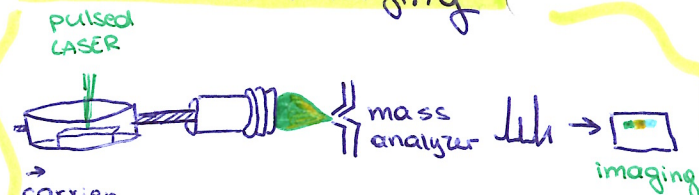
Interferences of ICP-OES

$\rightarrow$ Spectral Interference

overlap of signals

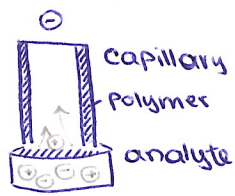
$\rightarrow$ change λ !

ICP - Laser Imaging

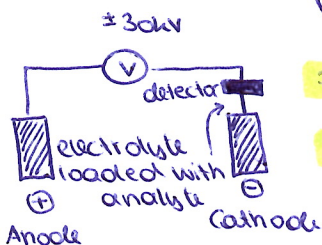


- a pulsed LASER evaporates material from the surface
- this is transported to the ICP torch by the carrier gas
- this allows to scan (image) surfaces
- by firing at the same spot = depth e.g. rat kidney, steel pipe shows Cu or Pt intensity!

CZE Capillary Zone Electrophoresis:



- 1) sample is labelled (fluorescence)
- 2) analyte is loaded into the capillary electrokinetically by applying a voltage. (can load a huge number of particles!)



- 3) Ends of capillary are inserted into anode & cathode buffers
- 4) instrument controls temperature & voltage

5) fragments are separated by size (small = fast) and charge ratio (m/z)

6) Detection via UV-absorbance, conductivity or fluorescence

optimizing:

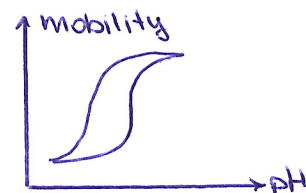
- change pH = changes degree of dissociation of the analytes or reverses charge
- addition of complex forming agents (to change charge or size)
- adding chiral selectors

Electro osmotic flow:

1) The silica gel has exposed silanol groups $\Rightarrow$ they are deprotonated at $pH \geq 3$

2) This leads to a buildup of a fixed layer of cations (inner Helmholtz layer)

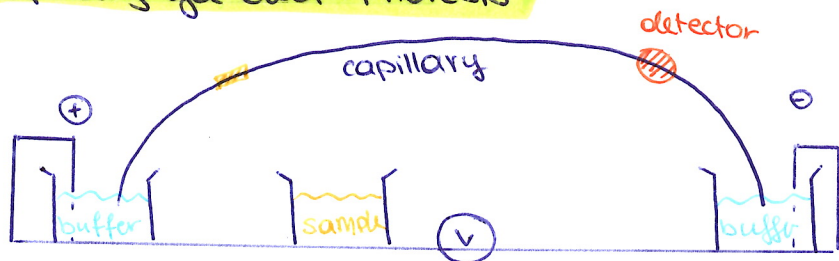
3) The outer Helmholtz layer (more + than -) then moves to $\ominus$ cathode & takes the whole electrolyte with it...



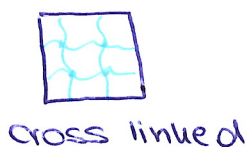
HOW TO PREVENT

- increase ionic strength of the electrolyte
- add org. solvents to the electrolyte
- add EOF modifiers

Capillary gel electrophoresis



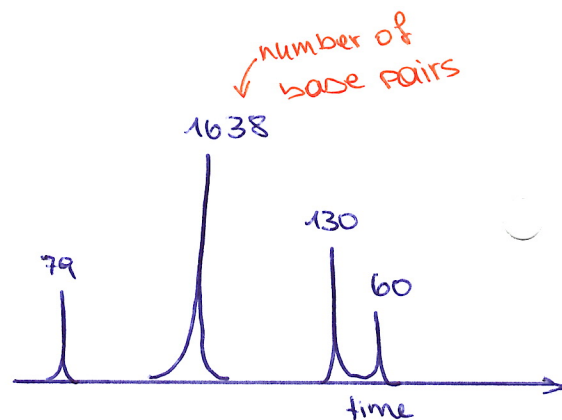
- 1) capillary is filled with buffered electrolyte in a gel (polyacrylamide gel)
- 2) analyte migrates through electrolyte solution under the influence of an electrical field
- 3) Separation according to size



↳ can be flushed & refilled

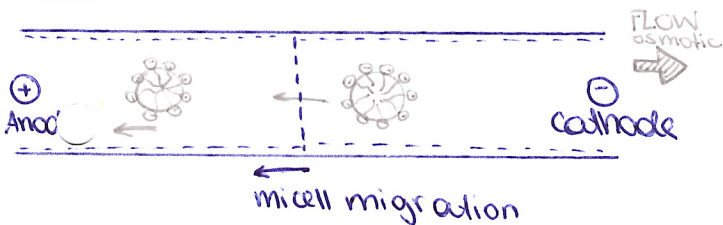
⇒ for proteins:

SDS = sodium dodecyl sulfate is used to form a negatively charged complex with proteins to keep the m/z constant, so that the analyte is only separated by size.



MEC Micellar electrokinetic chromatography

= same instrument as in CZE



1) carrier electrolyte contains micellar forming additives (SDS, sodium dodecyl sulfate)

2) SDS acts as a "stationary" phase (negatively charged)

3) separation according to partition between mobile & "stat." phase.

= suited for neutral analytes!

(hydrophobic analytes prefer the micell & will take long to reach the detector)

mobility

• acceleration

$$K_{acc} = z \cdot e \cdot E$$

$\begin{matrix} \nearrow \text{charge unit} \\ \searrow \text{voltage / length} \end{matrix}$
 $\begin{matrix} \nwarrow \text{charge number} \end{matrix}$

• friction / Stokes law:

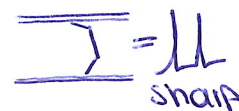
$$K_{fr} = 6\pi\eta r V_{sp}$$

$\begin{matrix} \nwarrow \text{viscosity} \\ \searrow \text{ion radius} \end{matrix}$
 $\nearrow \text{velocity}$

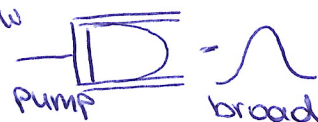
• steady state: $K_{fr} = K_{acc}$

flow profiles

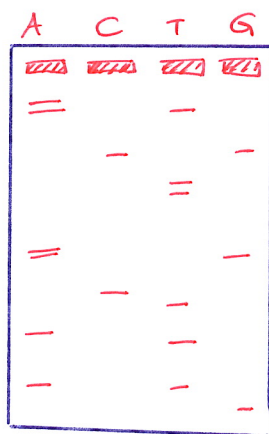
• electro osmotic flow



• hydrodynamic flow



Gel electrophoresis



• agarose or polyacrylamide on a glass plate

• voltage separated by m/z

• in SDS-PAGE: separation in 2 Dimensions

1) IEF: by isoelectric point

2) according to molecular weight

Electroblotting after gelelectrophoresis

1) transfers proteins/DNA onto a membrane for further analysis (e.g. antibodies, ligands...)

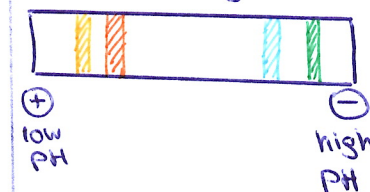


IEF = Isoelectric focusing

1) isoelectric point:

The pH where a molecule carries no net charge (=neutral)

2) INIEF the molecules are separated by their isoelectric points via a pH gradient & a voltage



at low pH = $\oplus$ charge proteins

at high pH = $\ominus$ charge proteins

With voltage they get pulled to the pH with their isoelectric point